AUXILIARY IDENTIFICATIONS FOR THE ISOMORPHISM PSEUDO TORSOR OVER A SPACE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of the isomorphism functor for a pair of objects. Invoked at the start of the counted proof (source line 1427). Gives part (1).
- The exact sequence relating the isomorphism functor to the group of automorphisms. Invoked inside the counted proof (source line 1429). Gives part (2) after etale localization.
- The corresponding statements for the pseudo torsor over a space. Invoked at the end of the counted proof (source line 1432). Reduces part (4) to part (2).

Source reference. Original definition definition-isom: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1357–1357. Navigation pointer only: the counted text refers to the Stacks item definition-isom, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original equation equation-exact-sequence-isom: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1331–1331. Navigation pointer only: the counted text refers to the Stacks item equation-exact-sequence-isom, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-isom-pseudo-torsor-aut: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 311–311. Navigation pointer only: the counted text refers to the Stacks item lemma-isom-pseudo-torsor-aut, which is an existing labeled result of this same chapter and is not part of

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this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let Z be an algebraic space and let $x_i : Z \to \mathcal{X}$, $i = 1, 2$ be morphisms. Then*

(1) *$Isom_{\mathcal{X}/\mathcal{Y}}(x_2, x_2)$ is a group algebraic space over Z,*
(2) *there is an exact sequence of groups*

$$0 \to Isom_{\mathcal{X}/\mathcal{Y}}(x_2, x_2) \to Isom_{\mathcal{X}}(x_2, x_2) \to Isom_{\mathcal{Y}}(f \circ x_2, f \circ x_2)$$

(3) *there is a map of algebraic spaces $Isom_{\mathcal{X}}(x_1, x_2) \to Isom_{\mathcal{Y}}(f \circ x_1, f \circ x_2)$ such that for any 2-morphism $\alpha : f \circ x_1 \to f \circ x_2$ we obtain a cartesian diagram*

$$\begin{array}{ccc} Isom^{\alpha}_{\mathcal{X}/\mathcal{Y}}(x_1, x_2) & \longrightarrow & Z \\ \downarrow & & \downarrow{\scriptstyle \alpha} \\ Isom_{\mathcal{X}}(x_1, x_2) & \longrightarrow & Isom_{\mathcal{Y}}(f \circ x_1, f \circ x_2) \end{array}$$

(4) *for any 2-morphism $\alpha : f \circ x_1 \to f \circ x_2$ the algebraic space $Isom^{\alpha}_{\mathcal{X}/\mathcal{Y}}(x_1, x_2)$ is a pseudo torsor for $Isom_{\mathcal{X}/\mathcal{Y}}(x_2, x_2)$ over Z.*

Proof. Part (1) follows from Definition definition-isom. Part (2) comes from the exact sequence (equation-exact-sequence-isom) étale locally on Z. Part (3) can be seen by unwinding the definitions. Locally on Z in the étale topology part (4) reduces to part (2) of Lemma lemma-isom-pseudo-torsor-aut. □